

TROMSO, Norway

WMO: 010260

Lat: 69.6536N

Lon: 18.9367E

Elev: 376

StdP: 14.50

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.3	(c) 13.9	(d) 2.1	(e) 6.2	(f) 15.2	(g) 5.3	(h) 7.3	(i) 16.1	(j) 26.4	(k) 30.3	(l) 24.1	(m) 30.1	(n) 4.8	(o) 200	(p) 0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	70.5	58.1	66.9	56.3	63.5	54.7	59.6	68.3	57.5	65.1	55.7	62.3	6.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.3	65.9	62.7	53.5	61.8	61.1	51.9	58.2	59.1	26.3	68.7	24.9	65.4	23.8	62.4	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.1	(b) 18.6	(c) 16.5	(e) 8.0	(f) 76.9	(g) 3.8	(h) 5.0	(i) 5.3	(j) 80.5	(k) 3.1	(l) 83.4	(m) 0.9	(n) 86.2	(o) -1.8	(p) 89.8		
(a) 21.1	(b) 18.6	(c) 16.5	(e) 8.0	(f) 76.9	(g) 3.8	(h) 5.0	(i) 5.3	(j) 80.5	(k) 3.1	(l) 83.4	(m) 0.9	(n) 86.2	(o) -1.8	(p) 89.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	25.8	26.3	28.6	34.9	42.8	48.2	54.6	52.8	46.6	38.0	31.9	29.3	
	DBStd	11.62	6.83	6.95	5.98	5.41	6.13	5.64	5.59	5.05	5.44	6.20	5.87	6.17	
	HDD50	4595	751	664	662	452	238	101	15	26	127	375	543	641	
	HDD65	9714	1216	1084	1127	902	687	506	325	379	551	838	993	1106	
	CDD50	361	0	0	0	0	16	45	159	113	26	2	0	0	
	CDD65	5	0	0	0	0	0	0	4	1	0	0	0	0	
	CDH74	41	0	0	0	0	1	4	30	6	0	0	0	0	
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0	
Wind	WSAvg	7.5	9.2	9.2	8.8	7.2	6.9	6.3	5.6	5.6	6.3	7.4	8.4	9.2	
Precipitation	PrecAvg	38.3	3.7	3.5	2.5	2.1	1.7	2.0	2.6	2.8	4.1	5.1	4.0	4.1	
	PrecMax	57.9	11.1	8.5	7.2	5.3	5.1	4.1	4.8	6.5	8.3	12.4	9.6	9.8	
	PrecMin	26.0	1.2	0.1	0.3	0.2	0.3	0.2	0.5	0.6	0.7	0.7	0.5	0.2	
	PrecStd	8.1	2.0	1.9	1.6	1.1	1.0	1.0	1.1	1.6	2.0	2.6	1.9	2.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	42.1	44.4	54.4	68.3	71.7	78.1	73.6	63.8	56.2	48.0	44.4	
		MCWB	37.8	37.5	37.0	44.2	52.9	55.2	62.7	59.8	53.9	49.3	43.2	39.6	
	2%	DB	39.8	40.2	41.3	48.6	61.5	65.8	72.2	68.1	60.2	51.8	44.2	41.9	
		MCWB	36.4	36.4	36.2	41.2	49.5	54.0	59.6	57.5	52.5	46.2	40.5	38.3	
	5%	DB	37.8	38.5	39.2	45.3	56.6	62.0	68.6	63.9	57.3	49.2	42.2	39.8	
		MCWB	34.7	35.3	35.2	39.2	46.8	52.1	57.8	56.0	50.8	44.9	38.5	36.2	
	10%	DB	35.5	36.4	37.3	42.6	52.7	58.0	65.1	61.0	54.7	46.7	40.3	37.9	
		MCWB	32.6	33.6	33.9	37.2	44.8	50.1	56.1	54.6	49.4	43.3	37.2	34.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	39.2	38.5	44.9	54.0	57.7	63.9	61.8	55.5	49.8	44.5	41.1
			MCDB	41.0	41.2	41.7	53.0	67.1	68.7	74.8	68.8	62.0	55.1	47.2	43.0
2%		WB	37.0	36.9	37.2	41.9	50.1	55.1	60.7	58.5	53.1	47.2	41.3	38.7	
		MCDB	39.3	39.4	40.5	47.5	60.4	64.0	71.0	65.6	58.3	51.0	43.5	41.6	
5%		WB	35.0	35.6	35.7	39.8	47.4	52.9	58.6	56.7	51.7	45.3	39.1	36.7	
		MCDB	37.4	38.1	39.0	44.6	55.6	60.4	67.1	62.8	56.2	48.5	41.7	39.3	
10%		WB	32.9	33.9	34.0	37.9	45.2	50.7	56.7	55.1	50.0	43.5	37.2	34.8	
		MCDB	35.1	36.3	37.1	42.0	51.8	57.2	64.0	60.5	54.0	46.6	39.8	37.4	
Mean Daily Temperature Range		5% DB	MDBR	6.6	7.2	8.1	9.9	11.0	10.7	12.3	11.4	9.4	6.9	6.1	6.7
			MCDBR	7.6	8.1	7.5	13.0	18.1	18.1	19.4	16.8	14.2	8.8	7.9	8.5
	MCWBR		6.4	6.3	5.5	7.6	9.3	9.3	9.6	8.8	7.9	6.3	6.0	6.7	
	5% WB	MCDBR	7.3	8.0	7.2	11.8	17.1	16.1	17.8	15.4	12.5	8.1	7.5	8.1	
		MCWBR	6.4	6.7	5.7	7.4	9.2	8.9	9.5	8.7	7.6	6.3	6.1	6.8	
Clear-Sky Solar Irradiance	taub	N/A	0.268	0.316	0.333	0.345	0.343	0.347	0.335	0.316	0.275	N/A	N/A		
	taud	N/A	2.051	2.113	2.227	2.381	2.466	2.488	2.506	2.513	2.197	N/A	N/A		
	Ebn at Noon	N/A	140	212	252	266	271	265	255	226	139	0	0		
	Edh at Noon	N/A	14	26	31	30	29	28	24	18	11	0	0		
All-Sky Solar Radiation	RadAvg	3	83	429	1008	1455	1524	1380	980	518	158	0	0		
	RadStd	0	17	59	97	96	197	143	81	60	25	0	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page